Voice Forgery Detection Using Constant Q Cepstral Coefficients (CQCC)

Todisco et al. (2017), in their paper *Constant Q Cepstral Coefficients: A Spoofing Countermeasure for Automatic Speaker Verification*, present a method for extracting acoustic features called Constant Q Cepstral Coefficients (CQCC), which combines spectral analysis by means of the Constant Q Transform (CQT) with traditional cepstral processing. The study focuses on the effect of the audio representation on the ability to detect voice manipulations, based on the hypothesis that a representation that differs from the spectral representations common in speaker verification systems may reveal artifacts that are not prominent in conventional features.

Unlike the Short-Time Fourier Transform (STFT), in which the frequency separation is uniform, CQT uses geometrically spaced frequency intervals and variable analysis windows, which allow a higher frequency resolution at low frequencies and a higher time resolution at high frequencies. In order to produce CQCC, the researchers developed a process that combines computing the power spectrum, applying a logarithm, resampling the spectrum onto a uniform frequency axis, and applying a Discrete Cosine Transform (DCT). This process makes it possible to convert the spectral representation into cepstral coefficients that serve as input to a classifier.

The method was evaluated by means of a Gaussian Mixture Model (GMM) with 512 components per class, on three datasets: ASVspoof 2015, AVspoof and RedDots Replayed, which include Speech Synthesis, Voice Conversion and Replay attacks. On the ASVspoof 2015 dataset an average EER of about 0.26% was achieved over all attacks and 0.46% on unknown attacks, where the latter result represents a relative improvement of about 72% compared with the previous system with the best performance. On the AVspoof dataset an HTER of 0.67% was achieved, and on the RedDots Replayed dataset an EER of 1.85% was achieved, results that reflect an improvement relative to the previous comparison methods.

However, the Cross-Database evaluation demonstrated that the method's performance is sensitive to the choice of the feature configuration and to the attack scenario. Whereas in AVspoof a combination of static and dynamic features achieved the best result, in RedDots Replayed the optimal combination included Delta and Acceleration features without static features. In addition, despite the high overall performance in AVspoof, a certain unknown Replay attack showed a significantly high error rate. These findings emphasize the generalization limitations of a single feature configuration and the importance of evaluating performance also by attack type and not only by an overall metric.

The central contribution of the study is in demonstrating the potential of improving the audio representation, even when a classical statistical classifier is used. However, since the study was conducted on earlier datasets and forgery methods, one should not conclude directly about the effectiveness of CQCC against modern technologies for generating synthetic speech. For the present study, CQCC constitutes a relevant alternative to MFCC and makes it possible to examine, in a controlled manner, the contribution of the audio representation to forgery detection performance. Accordingly, it is possible to compare MFCC with CQCC using the same classifier and evaluation protocol, and also to examine the effect of static and dynamic features on the EER and on the generalization ability across data sources and forgery methods.

Bibliographic source:

Todisco, M., Delgado, H., & Evans, N. (2017). *Constant Q Cepstral Coefficients: A Spoofing Countermeasure for Automatic Speaker Verification*. Preprint version, May 12, 2017.